

PRAMAAN

प्रमाण · proof

Turning geo-tagged watershed photographs
from paperwork into proof.

Ministry of Rural Development · Department of Land Resources

Category: Software · Theme: Agriculture, FoodTech & Rural Development

FIELD PHOTOGRAPH

“Check dam, completed,
20-Nov-2023, 19.87°N 77.42°E”

SATELLITE @ 30 m

MNDWI -0.08 → +0.31
Water 4 → 7 months / year

RECONCILE

CORROBORATED · L4 control-differenced · confidence 0.84

...and the evidence against it, shown alongside.

India solved collection. It has not solved interpretation.

1 crore+

MGNREGA assets geo-tagged
PIB, 2018

1,24,830

water harvesting structures created
or rejuvenated under WDC-PMKSY 2.0

₹8,134 cr

central outlay · 49.5 lakh ha target
2021–2026

Today the photograph is moderated by eye.

- NRSC's SRISHTI portal colour-codes each geotag blue → green (accepted) or red (rejected)
- An officer opens one photograph and clicks one of two buttons
- That verifies a photograph exists. It does not verify that the structure works.

ACCEPTED

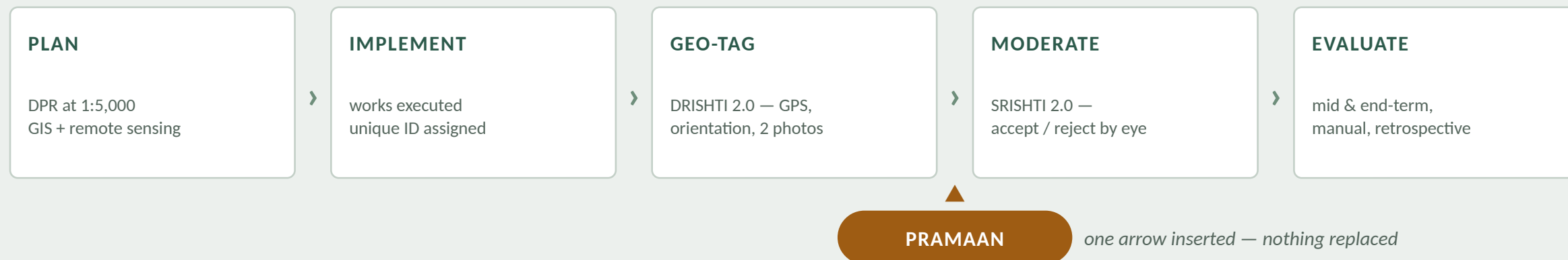
REJECTED

...and nothing in between.

No outcome. No confidence. No evidence.
No trend. No record of why.

1,24,830 structures is far past the point
where an officer can eyeball an outcome.

DoLR already has DRISHTI, SRISHTI and Bhuvan. We add the one missing layer.



The WDC-PMKSY 2.0 Guidelines already mandate this analysis:

“change detection technique should be used to understand changes in vegetation cover by NDVI and changes in water availability through NDWI” · “cross verification of satellite-based images with ground-based interventions can facilitate multiple levels of authentication, monitoring”

Nobody has built the system that does those two sentences at scale. That is PRAMAAN.

Every photograph becomes a testable claim.



0.15

PHOTO

What the field says. Weighted lowest — it is the claim’s own source.

0.25

TERRAIN

Flow accumulation, Strahler order, slope. Deterministic. No AI.

0.25

SATELLITE

NDVI • MNDWI • bare soil • water persistence, at 30 m.

0.25

TEMPORAL

Same season, year on year, against 5–12 matched controls.

0.10

CONTEXT

Rainfall anomaly, land cover, neighbouring interventions.

VERDICT

L4 • CONTROL-DIFFERENCED

confidence 0.84 coverage 0.92

DISSENT — ALWAYS SHOWN

- Cloud gap Jul–Aug 2024: 2 of 6 scenes lost
- Only one fully observed post-monsoon season
- Correlational — this is not a causal claim

The officer decides.

AI never becomes government evidence on its own.

Illustrative verdict card. Five independent evidence families; the weights are published in the interface and adjustable per deployment.

One workflow, built completely — not ten built partially.

01 SELECT

Open a real micro-watershed. 142 geo-tagged works, 6 already flagged.

02 INGEST

EXIF → sidecar → CSV → manual pin. Every coordinate carries its provenance rank.

03 INTERPRET

Photo attributes with calibrated confidence — and permission to abstain.

04 RECONCILE

Terrain, satellite, temporal and matched controls resolve to one verdict.

05 ADJUDICATE

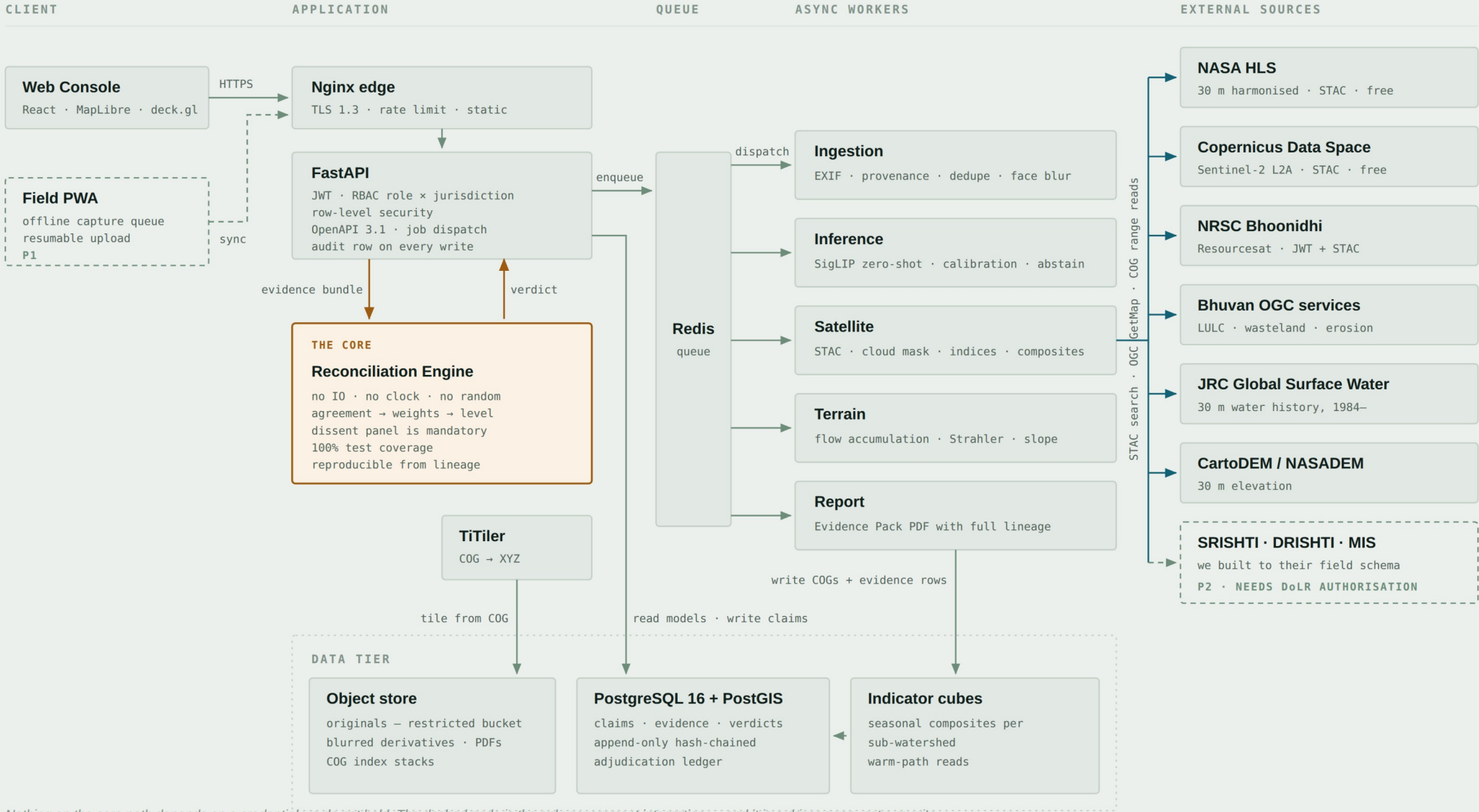
Accept, edit or reject. Signed into an append-only, hash-chained ledger.

06 REPORT

Evidence Pack PDF: every scene id, model version and weight used.

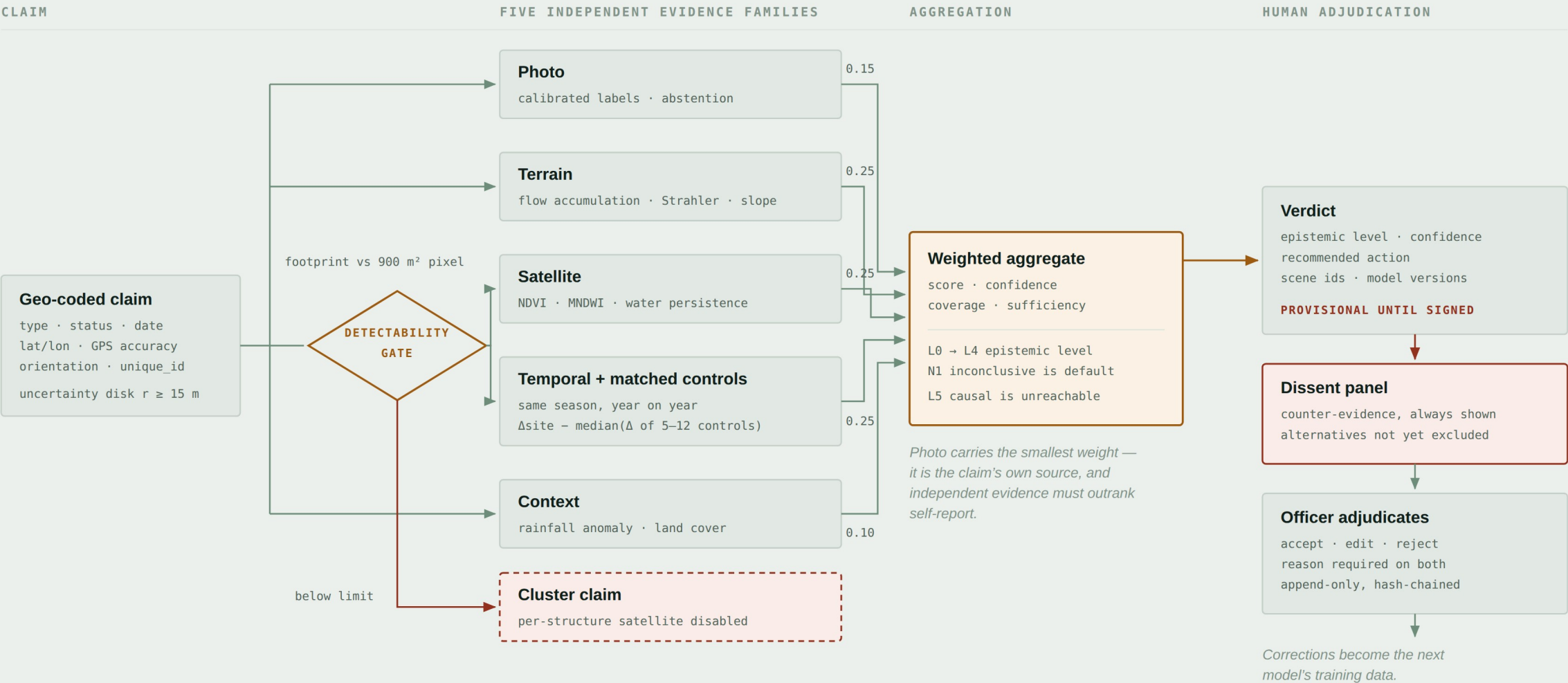
Boring where it should be. Careful where it matters.

The amber core has exactly two connections — a bundle in, a verdict out.



Nothing on the core path depends on a credential we do not hold. The dashed node is the only government integration — and it is a driver swap, not a rewrite.

Inside the engine — and the one branch that removes an arrow.



A 25 × 25 m farm pond is 625 m². One 30 m pixel is 900 m². The gate fails, and per-structure satellite evidence is switched off — visibly, in the UI and in the PDF.

Three ideas do the work. None of them is deep learning.

01

MATCHED CONTROLS

Every structure is scored against 5–12 automatically matched sites in the same sub-watershed — same slope, land cover, elevation and stream order, with no intervention.

Same rainfall, same season. We measure the intervention, not the monsoon.

02

TERRAIN PLAUSIBILITY

A check dam claimed on a ridge with 12 cells of flow accumulation is flagged before any model runs, from DEM-derived flow accumulation, Strahler order and slope.

Deterministic. Zero training data. Explainable to an auditor in one line.

03

THE DETECTABILITY GATE

Footprint is compared to sensor resolution first. Below one pixel, per-structure satellite evidence is disabled and the claim escalates to a cluster.

A system that refuses to answer is one you can trust when it does.

One real claim, run end to end on real data.

INCONCLUSIVE · N1

confidence 0.0615

Check dam · claimed complete 20-Nov-2023 · Marathwada · REAL DATA

- Satellite: 40 cloud-screened NASA HLS granules · 10 seasonal composites over 5 years · rabi NDVI +0.1157 across the claim date
- Controls: 12 sites matched on real NASADEM terrain rose a median +0.0901. The site sits at the 75th percentile — inside the band. Differenced +0.026
- Terrain: Strahler order 0, 277 m from any drainage line — implausible siting for a check dam. The engine names the conflict instead of averaging it

REQUIRES VERIFICATION · N3

confidence 0.3371

Farm pond · golden case 21 · SYNTHETIC INPUTS, ENGINE-COMPUTED

- Terrain: 12 cells of flow accumulation, 340 m from any channel — no runoff reaches it. A deterministic rule, not a model, drives this verdict
- Detectability gate FAILED: a 625 m² footprint against a 900 m² pixel. Per-structure satellite evidence is disabled, and the verdict is reached via the named N3_TERRAIN_PATH

MEASURED, NOT CLAIMED — every figure below was measured on this build and is re-runnable from the repository. NOT MEASURED YET: adjudication-time A/B test · photo-model precision/recall · calibration error (ECE) · terrain-screen precision. Where a number does not exist, this deck says so.

Engine throughput	~13 μ s / verdict	1,24,830 structures in under 2 s · one district of 1,200 works in 16 ms
Test suite	458 tests · 100 %	branch coverage on the deterministic core · 23 golden cases, all 8 levels
Matched controls	12 of 342	on real DEM covariates; 276 rejected for stream distance, 11 for slope
Imagery volume	1.8 GB	five years, windowed COG reads — against 78 GB for naive granule download
Photo inference	29 img/s on CPU	zero GPU · ~6 img/s on a demo VM assumed 5× slower · 1,200 images in minutes

What is genuinely new — and what is not.

NOT NEW — AND WE DON'T PRETEND OTHERWISE

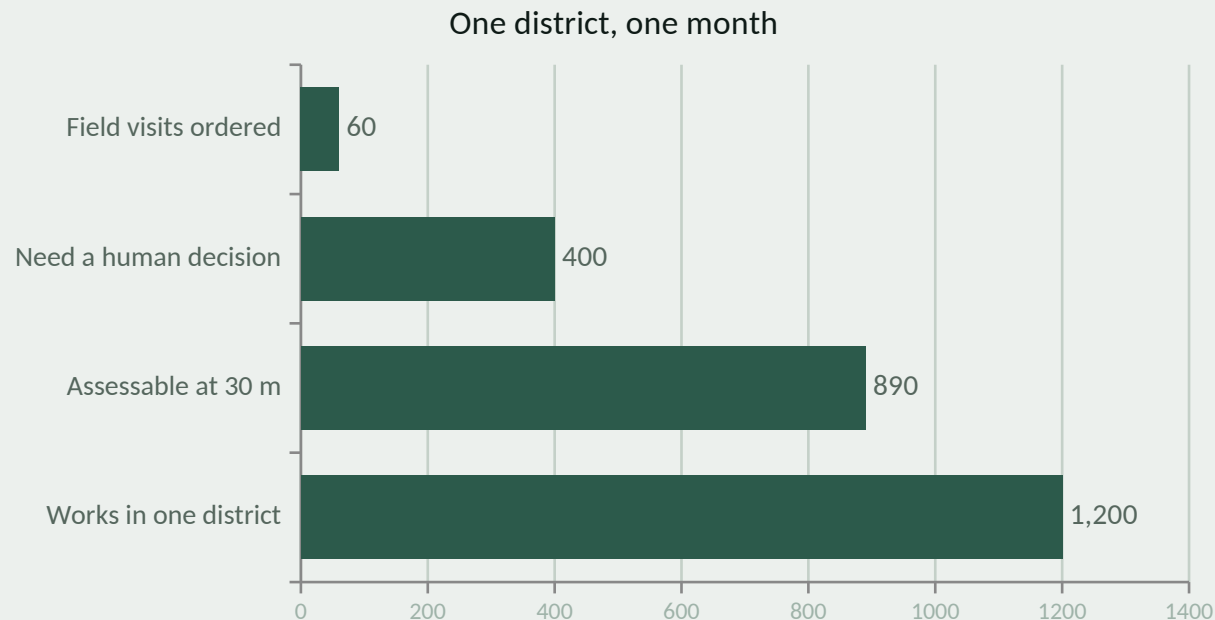
- NDVI / NDWI change detection
- Mapping geo-tagged points on satellite imagery
- GIS dashboards for government programmes
- Computer vision applied to field photographs
- Treated-versus-control watershed impact studies

NEW IN THIS COMBINATION

- Claim-level satellite-field reconciliation on a published epistemic ladder
- Matched-control differencing per individual structure, not per programme
- Terrain plausibility screening of geotags before any model runs
- An explicit refusal-to-answer gate tied to sensor resolution
- An append-only adjudication ledger that becomes the next training set

We are not claiming a first. We are claiming these five have not been assembled into one government-workflow-native system — and we checked.

Redirecting scarce human attention.



Illustrative funnel from the demo corpus. The ratio is measured and reported — never assumed.

Watershed development already works.

B:C $\approx$ 2 · IRR 27.4 % · cropping intensity +35.5 %
runoff -45 % · soil loss -1.1 t / ha / yr

ICRISAT meta-analysis, 636 Indian micro-watersheds

The constraint is not whether it works. It is knowing which structures are working.

₹0

in data and licence costs. Every source on the core path is free, open and verified — so the marginal cost of monitoring one more structure is CPU seconds.

Every officer correction becomes a labelled training sample — the dataset is produced by the system's own use, by domain experts, for free, forever.

Free data. No GPU. One driver class to production.

What production needs	What we use for the hackathon	Swap effort
DRISHTI geo-tag feed	A corpus faithful to NRSC’s published DRISHTI field schema — lat, lon, GPS accuracy, orientation, timestamp, activity, survey no., status	one driver class
“SRISHTI-DRISHTI 30 m imagery”	NASA HLS 30 m + Landsat 30 m + Resourcesat through the NRSC Bhoonidhi STAC API	one driver class
WDC-PMKSY MIS records	CSV import matching the MIS columns, keyed on the government’s own unique_id	configuration only
SRISHTI project boundaries	SLUSI micro-watershed polygons — already the same geometry family	none

VERIFIED, OPEN, PROGRAMMATIC

NASA HLS 30 m (STAC) · Copernicus Sentinel-2 (STAC) · NRSC Bhoonidhi API (JWT + STAC; open above 5 m under Indian Space Policy 2023)
Bhuvan OGC WMS · JRC Global Surface Water · CartoDEM

WE COULD NOT FIND A PUBLIC SRISHTI-DRISHTI API

So we are not going to pretend we did. We built to NRSC’s published field schema instead — which makes integration a configuration change on the day the Department grants access, not a rewrite.

The officer decides. The system shows its work.

ADJUDICATION

ACCEPT

EDIT

REJECT

SIGNED · R. Kumar, WDC Project Manager, Nanded
27-Aug-2026 09:14 IST · ledger hash 7f3a...9c1

Append-only · hash-chained · non-repudiable
Every correction becomes a labelled sample for the next model.

*Nothing here becomes government evidence
until a named human being says it does.*

**PRAMAAN is the intelligence layer
DoLR's own guidelines already asked
for.**

- Integrates — never replaces SRISHTI, DRISHTI or Bhuvan
- Implements two written mandates from WDC-PMKSY 2.0
- Adopts the government's own IDs, roles and statuses
- Caps itself below causal claims — enforced in code
- Runs on free, verified data with zero licence cost

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1,24,830 structures. Free satellite data. The officer still decides.